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1  //##### LESSON 2 - CONTROL STRUCTURES #####
2  /*THE FOUNDATION OF ALL LOGIC IS THE if STATEMENT*/
3  var firstName = "Sean";
4
5  /*IT STARTS WITH THE KEYWORD if, FOLLOWED BY PARENTHESES. WITHIN THE PARENTHESES IS
   THE CONDITION TO BE CHECKED FOR "TRUTHINESS". IF THE CONDITION IS TRUE, THE CODE
   ENCLOSED IN THE ATTACHED CURLY BRACES WILL EXECUTE; IF NOT, NOTHING WILL HAPPEN.*/
6  if(firstName == "Sean"){ //Check for NOT equal to with !==
7      alert("Hello Sean!");
8  }
9  //NOTE THE DOUBLE EQUALS SIGN. THAT IS BECAUSE WE ARE COMPARING THE TWO VALUES.
10 //OTHER COMPARISON OPERATORS:  >  <  >=  <=  ===
11 var myAge = 39;
12 if (myAge < 65){
13     alert("Not ready to retire yet.");
14 }
15
16 //IN JAVASCRIPT YOU MUST BEWARE OF TYPE COERCION. TO AVOID THIS ALWAYS USE === OR !==
   BECAUSE IT CHECKS FOR DATA TYPE.
17 var loggedIn = 0; //var loggedIn = "";
18 if (loggedIn == false){
19     alert("That can't be true!");
20 }
21 //FOR MULTIPLE EVALUATIONS, WE ALSO HAVE LOGICAL OPERATORS...
22 //  && (Logical "and")  || (Logical "or")
23 if (firstName === "Sean" || firstName === "Bernie"){
24     alert("Hello Sean or Bernie");
25 }
26 var lastName = "Doyle";
27 if (firstName === "Sean" && lastName === "Doyle"){alert("Hi Sean Doyle");}
28
29 //===== confirm() =====
30 //THE confirm FUNCTION ALLOWS US TO GET A YES OR NO ANSWER FROM OUR USER
31 //YOU NEED TO CREATE A VARIABLE TO HOLD THE USER'S RESPONSE
32 var userInput = confirm("Are you having fun yet?");
33 console.log(userInput); //NOTE THE VALUES THAT IT RETURNS
34
35 //NOW, OUR SIMPLE if STATEMENT ISN'T QUITE SUFFICIENT.
36 //THE ANSWER IS THE else STATEMENT
37 if(userInput === true){
38     alert("You said Yes!");
39 } else {
40     alert("You did not say Yes.");
41 }
42 //FOR MORE THAN TWO OPTIONS, WE HAVE THE else if STATEMENT
43 //WE CAN HAVE MULTIPLE else if'S BUT IT SHOULD END WITH AN else
44 if (firstName === "Bernie"){
45     alert("Hi Bernie!");
46 } else if(firstName === "Sean"){
47     alert("Hi Sean");
48 } else {
49     alert("Sorry, I don't know you.");
50 }
51
52 //===== prompt() =====
53 //THE prompt FUNCTION ALLOWS US TO GET A STRING FROM OUR USER.
54 var fName = prompt("Please log in...");
55 console.log(fName); //WHAT IF YOU ENTER A NUMBER?
56
57 var luckyNum = prompt("What is your lucky number?");
58 //TO HANDLE NUMBERS, THERE ARE TWO USEFUL TOOLS...
59 //isNaN() IS A BUILT IN FUNCTION THAT CHECKS IF A VALUE isNot a Number.
60 if (isNaN(luckyNum)){alert("Hey, that's not a number!");}
61 console.log(typeof luckyNum); //This will show the data type of luckyNum
62
63 //parseInt() IS A BUILT IN FUNCTION THAT MAKES A STRING A NUMBER

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64  var stringToNum = parseInt(luckyNum);
65
66  var fName = prompt("Please log in...", "First name only");
67  //prompt ALLOWS FOR AN OPTIONAL 2ND PARAMETER THAT PREPOPULATES THE TEXTBOX
68
69  //NOW, THE POSSIBLE INPUT IS ALMOST INFINITE. HOW DO WE MANAGE THIS?
70  //VALIDATION
71  //===== switch =====
72  //THE SWITCH ALLOWS US TO HANDLE MULTIPLE KNOWN VALUES, AS WELL AS SET A DEFAULT FOR
  EVERYTHING ELSE
73  var stopLight = prompt("What colour is the light?", "Red, Yellow, or Green");
74
75  switch(stopLight){
76      case "Green":
77          alert("Go!");
78          break;
79      case "Yellow":
80          alert("Clear intersection!");
81          break;
82      case "Red":
83          alert("Stop!!");
84          break;
85      default:
86          alert("Drive with caution!");
87          break;
88  }
89  /*THE PROGRAM RUNS THROUGH THE CASES STARTING FROM THE TOP. IF IT FINDS A MATCH, IT
  RUNS THE SUBSEQUENT CODE. IF NOT, IT WILL KEEP TRICKLING DOWN AND EXECUTE default.
  THE break STATEMENT TELLS THE PROGRAM TO EXIT THE switch ONCE IT HAS COMPLETED THE
  CORRECT CODE. NOTE WHAT HAPPENS IF YOU REMOVE THE break STATEMENTS.*/
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1 //CONTROLS PART 2 ===== switch NESTED WITHIN AN else STATEMENT =====
2
3 var userInput = prompt("Do you prefer red, green or blue?", "Please say blue.");
4 //USE console.log(userInput) TO SEE THE POSSIBLE VALUES THAT IT MIGHT RETURN.
5
6 //TO HANDLE THE INPUT, WE WILL FIRST USE AN if STATEMENT TO CHECK FOR INVALID DATA, AND
  SEND AN ALERT MESSAGE IF WE RECEIVE INVALID INPUT. NOTE THE 3 POSSIBILITIES FOR EMPTY
  DATA: AN EMPTY STRING; A null VALUE (IF THE USER CLICKS CANCEL); AND THE DEFAULT TEXT
  THAT WE CREATED.
7
8 //IF THE if STATEMENT DOES NOT FIND INVALID DATA, THE CODE IN THE else BLOCK WILL
  EXECUTE. NESTED INSIDE THE else BLOCK WE USE A switch STATEMENT TO LOOK FOR OUR
  EXPECTED VALUES.
9
10 //HERE, WE ARE VALIDATING OUR DATA BY CHECKING FOR WHAT WE DON'T WANT FIRST
11 //WE NEED TO CHECK FOR AN EMPTY STRING, NULL, AND DEFAULT TEXT
12 if(userInput === "" || userInput === null || userInput === "Please say blue."){
13     alert("You need to enter a colour!");
14
15 } else {
16     //NOW WE CAN CHECK OUR VALID(ISH) DATA FOR OUR KNOWN CASES
17     switch(userInput.toLowerCase()){
18         case "red":
19             alert("Red is rad");
20             break;
21
22         case "green":
23             alert("Green is great!");
24             break;
25
26         case "blue":
27             alert("Blue is best!");
28             break;
29
30         default:
31             alert("Not one of the big 3");
32             break; //break ISN'T NECESSARY HERE, BUT IT'S BEST PRACTICE
33     }
34 } //END else STATEMENT
35
36
37 /*===== SOME MORE USEFUL TOOLS FOR YOUR TOOLBOX =====
38 console.log();//output messages to the console
39 console.log("check value" + myValue);
40
41 .toLowerCase(); //converts a string to all lowercase letters
42 myString.toLowerCase(); //"HeLlo" would now equal "hello"
43
44 parseInt() //converts a string number into a number IF it can
45 parseInt(myNum); //"56" would become the number 56
46
47 typeof // returns the data type of a value
48 typeof "56" //returns "string" -note 56 is in quotes therefore a string!
49
50 isNaN //checks if a value is Not a Number and returns true or false
51 isNaN("Hi"); //returns true. "Hi" is Not a Number.
52
53 For more detailed information about how these (and other functions) work, the Mozilla
  Developer Network (MDN) is a great resource: https://developer.mozilla.org
54 */

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